

A Large Language Model Analysis of Media and Congressional Discourse on the ESA

1 Chapter 1 - Methods: Extracting ESA Narratives

A Comparison of LLM-Based NPF Extraction and Structural Topic Modelling

1.1 Text corpora

We assembled two text corpora. The news corpus consists of 300 *Wall Street Journal* (WSJ) articles identified through the ProQuest database using a keyword search procedure targeting ESA listing events. Articles were retrieved for the period 1984–2025. The congressional corpus consists of 300 documents from the Congressional Record retrieved via the GovInfo API for the period 1995–2025, including floor speeches, committee statements, and extensions of remarks referencing ESA listing events. Beginning the congressional corpus in 1995 reflects the availability of systematic digital records through GovInfo, and is consistent with the period following the 104th Congress when ESA oversight activity intensified (Repetto, 2006).

Put a figure

Both corpora were stratified by ESA policy category to ensure representation across the range of ESA policy activities. Ten meta-categories were constructed: ESA Regulatory/Agency Action (news: 19.3%, congress: 19.0%), ESA Legislative Action (news: 17.3%, congress: 42.7%), ESA Species Listing (news: 17.3%, congress: 5.7%), ESA Litigation (news: 11.0%, congress: 1.0%), ESA Habitat/Recovery (news: 9.7%, congress: 7.3%), ESA Judicial Review (news: 9.3%, congress: 1.0%), ESA Species Delisting (news: 4.3%, congress: 5.0%), ESA Conservation Agreements (news: 1.7%, congress: 0.7%), ESA Funding (news: 1.7%, congress: 9.0%), and

Other/Non-ESA (news: 8.3%, congress: 8.7%). The disparity in Species Listing representation between corpora — 17.3% in news versus 5.7% in congress — is itself substantively informative and consistent with the agenda-setting hypothesis.

1.2 Data Collection

1.3 Outcome and predictor variables

The primary analysis focuses on ESA Species Listing documents, the category with the largest disparity between news and congressional representation and therefore the most direct test of the agenda-setting hypothesis for this specific policy process.

The ESA encompasses a wide range of policy activities, but species listing is among the most prominent and recurring topics in ESA discourse — it is the foundational act that triggers federal protections, generates public controversy, and consistently attracts media attention from diverse stakeholders (Repetto, 2006). Despite this centrality, listing appears in only 5.7% of congressional documents compared to 17.3% of news articles in our corpus. This three-to-one disparity between media coverage and congressional production is precisely the structural gap that agenda-setting theory predicts, and it makes listing the most direct and interpretable test of the hypothesis. The primary analysis therefore focuses on ESA Species Listing.

The **outcome variable** is $cong_t$: the count of congressional ESA listing documents produced in month t , aggregated from the 300-document congressional corpus. Monthly counts were constructed by assigning each document to its date of publication and summing within calendar months over the period January 1995 to December 2025, yielding $n = 371$ monthly observations.

The **primary predictor** is $news_lag$: the count

of WSJ ESA listing articles published in month $t-1$. The one-month lag reflects the theoretical prediction of the agenda-setting hypothesis that media coverage precedes and influences legislative attention — and ensures that the causal direction is identified by time ordering (Granger, 1969).

The **control variable** is $cong_lag$: the count of congressional ESA listing documents in month $t-1$, included to capture congressional own-momentum and separate it from the media effect. Without this control, the news coefficient would absorb autocorrelation in congressional activity that is unrelated to media influence.

2 Statistical Approach

Selecting the appropriate model for $cong_t$ requires first characterising the properties of the outcome variable, then evaluating each candidate model against those properties. We follow the protocol of making model selection decisions on structural grounds — based on the nature of the data generating process — rather than on post-hoc fit comparisons (Zuur et al., 2010; Gelman and Hill, 2007).

2.1 Properties of the outcome variable

Table 1 presents summary statistics for $cong_t$ across the 371 monthly observations.

Four properties of $cong_t$ are immediately relevant to model selection. First, the outcome is a *count*: it can only take non-negative integer values. Second, it is *heavily zero-inflated*: 96.8% of observations are zero, substantially more than would be expected from a Poisson distribution with the observed mean (Welsh et al., 2000; Mullahy, 1986). Third, it is *overdispersed*: the variance-to-mean ratio of 2.25 violates the Poisson assumption that variance equals mean (Cameron and Trivedi, 1998). Fourth, it exhibits *rare episodic spikes*: the maximum of 5 documents in January 2025 represents a value more than 100 times the overall mean, and the distribution is correspondingly skewed (*skewness* = 11.27, *kurtosis* = 157.93).

These four properties jointly eliminate Ordinary Least Squares as a candidate model and progressively constrain the choice toward the Hurdle model, as we now show.

2.2 Why ordinary least squares is inappropriate

OLS models the expected outcome as a linear function of predictors, minimising the sum of squared

residuals:

$$cong_t = \alpha + \beta_4 \cdot news_lag + \beta_3 \cdot cong_lag + \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, \sigma^2) \quad (1)$$

Three structural requirements of OLS are violated by $cong_t$. First, OLS assumes a *continuous normally distributed outcome*, but $cong_t$ is discrete and bounded below at zero. For April 1999, where $news_lag = 3$ and the actual outcome was $cong_t = 1$, OLS predicts:

$$\widehat{cong}_t = 0.038 + 0.068 \times 3 + 0.033 \times 0 = 0.242$$

This prediction is categorically impossible. No data-generating process for whole-number counts can produce 0.242 documents. Second, OLS assumes *constant variance* (homoscedasticity) in the residuals, but count data variance typically grows with the mean, producing the wedge-shaped residual pattern characteristic of heteroscedastic count data (Cameron and Trivedi, 1998). Third, the fitted regression line can produce *negative predictions* for months with low $news_lag$ values, which are impossible for counts. OLS is therefore

2.3 Poisson regression: the correct family, with a violated assumption

Poisson regression belongs to the Generalised Linear Model family (Nelder and Wedderburn, 1972; McCullagh and Nelder, 1989) and is the standard baseline model for count data. It models the expected count through a log link function:

$$\log(\lambda_i) = \alpha + \beta_4 \cdot news_lag_i + \beta_3 \cdot cong_lag_i \\ \lambda_i = \exp(\alpha + \beta_4 \cdot news_lag_i + \beta_3 \cdot cong_lag_i)$$

The exponential function guarantees $\lambda_i > 0$ for all covariate values, resolving the impossibility problem of OLS. For April 1999, Poisson gives $\hat{\lambda} = 0.336$, and the resulting Poisson distribution assigns $P(cong_t = 1) = 24.0\%$ — meaning the actual observed outcome falls within a meaningful probability region of the model’s predictions.

Poisson regression also confirmed the direction of the agenda-setting signal: $\hat{\beta}_4 = 0.714$, corresponding to a rate ratio of $\exp(0.714) = 2.04$. Each additional WSJ ESA listing article in the previous month is associated with a doubling of the expected congressional count in the current month, holding prior congressional activity constant.

However, the Poisson distribution requires that the variance equal the mean — a property known as equidispersion:

$$Var(cong_t) = \lambda \quad (2)$$

For $cong_t$, the variance (0.103) exceeds the mean (0.046) by a factor of 2.25, confirming *overdispersion* (Cameron and Trivedi, 1998; Hilbe, 2011). The consequence of fitting Poisson to overdispersed data is systematic underestimation of standard errors by a factor of approximately $\sqrt{2.25} = 1.50$, producing p-values that are too small and inflated Type I error rates (Ver Hoef and Boveng, 2007). Standard Poisson is therefore insufficient for this data, though it establishes the correct model family and baseline coefficient direction.

2.4 Quasi-Poisson: correcting the standard errors

Quasi-Poisson (McCullagh and Nelder, 1989) relaxes equidispersion by introducing a dispersion parameter φ , estimated from the Pearson residuals:

$$\hat{\varphi} = \frac{1}{n-p} \sum_{i=1}^n \frac{(cong_{t_i} - \hat{\lambda}_i)^2}{\hat{\lambda}_i} = \frac{919.80}{368} = 2.50 \quad (3)$$

where $n = 371$ and $p = 3$ (α, β_4, β_3). The variance is modelled as $Var(cong_t) = \varphi \cdot \lambda$, and all standard errors are multiplied by $\sqrt{\hat{\varphi}} = \sqrt{2.50} = 1.581$. This correction preserves the Poisson coefficient estimates while producing honest uncertainty quantification.

Quasi-Poisson is a correction applied to the Poisson likelihood after fitting, not a re-estimation under a new likelihood. The coefficients are therefore identical to Poisson. Quasi-Poisson treats the variance as growing linearly with the mean (equation 4):

$$Var(cong_t) = \varphi \cdot \lambda \quad (4)$$

While Quasi-Poisson addresses the standard error problem, it does not address the structural nature of the zero observations, as we discuss in Section 2.6.

2.5 Negative Binomial regression: quadratic overdispersion and fat tails

The Negative Binomial (NB) distribution (Hilbe, 2011) provides a structurally richer model for overdispersion than Quasi-Poisson. It introduces a dispersion parameter θ estimated through maximum likelihood, and models the variance as a quadratic function of the mean:

$$Var(cong_t) = \mu + \frac{\mu^2}{\theta} \quad (5)$$

The μ^2/θ term grows faster than the mean as μ increases, producing fat tails that assign non-trivial probability to extreme counts. When $\theta \rightarrow \infty$, equation `eq:nb_varcollapsestoPoisson` *equidispersion*; as $\theta \rightarrow 0$, the variance grows without bound, accommodating arbitrarily heavy tails. For $cong_t$, $\hat{\theta} = 0.058$ — a small value, consistent with the extreme January 2025 spike ($cong_t = 5$) being a genuine feature of the data-generating process rather than an outlier.

Unlike Quasi-Poisson, NB is a full re-estimation under the NB likelihood. Consequently, the regression coefficients differ from Poisson: $\hat{\beta}_4 = 0.486$ under NB (rate ratio = 1.63) compared to 0.714 under Poisson (rate ratio = 2.04). The difference reflects NB's absorption of extreme month variance into $\hat{\theta}$ rather than into the predictor coefficient estimates.

The choice between Quasi-Poisson and Negative Binomial is a structural one, not a numerical one. The Quasi-Poisson dispersion parameter φ and the NB parameter θ are not comparable quantities — they operate through different likelihoods on different mathematical scales. The structural argument for NB over Quasi-Poisson in this application is that the quadratic variance function in equation `eq:nb_varbetterrepresentsadata` — *generating process where extreme months — driven by concentrated political events — can occur with non-trivial probability that a linear variance model cannot assign.*

2.6 The zero structure: structural versus sampling zeros

Both Quasi-Poisson and Negative Binomial treat the 359 zero observations as draws from a single distribution with a low mean. However, 96.8% zeros in a monthly political activity series is not simply a consequence of a small expected count. It reflects a distinction that is critical for model specification (Mullahy, 1986; Welsh et al., 2000):

Sampling zeros arise when the expected count is low and the random draw from the count distribution happens to produce zero. These zeros are part of the count process and are appropriately modelled by Poisson or NB.

Structural zeros arise when congressional engagement with ESA listing does not occur in a given month because the issue has not crossed the attention threshold — regardless of news

coverage. These zeros come from a separate threshold process and cannot be appropriately modelled by the count distribution alone.

The distinction is theoretically motivated. As Baumgartner and Jones (1993) and Jones and Baumgartner (2004) argue, legislative attention to niche policy issues operates as a punctuated equilibrium: long periods of equilibrium (structural silence) are punctuated by episodes of intense attention driven by exogenous shocks. ESA listing is precisely this type of niche issue. The 12 active months in our dataset cluster around identifiable political events — the 1999 salmon crisis, the 2015 ESA reauthorisation debates, and the 2025 legislative push — consistent with a threshold process rather than a continuous low-probability one.

When a model treats structural zeros and sampling zeros as the same type of observation, the coefficient estimates represent a mixture of two different relationships: the relationship between news coverage and structural silence, and the relationship between news coverage and active count intensity. These are not the same relationship and should not be estimated as if they were.

2.7 The Hurdle model: separating silence from activity

The Hurdle model (Mullahy, 1986; Zeileis et al., 2008) explicitly represents the two-process structure described above. It consists of two components estimated simultaneously.

Part 1 (binary hurdle) models the probability that $cong_t$ crosses zero — whether ESA listing crosses the congressional attention threshold in month t :

$$P(cong_t > 0) = \text{logistic}(\alpha_0 + \beta_{40} \cdot \text{news_lag} + \beta_{30} \cdot \text{cong_lag}) \quad (6)$$

where $\eta = \alpha_0 + \beta_{40} \cdot \text{news_lag} + \beta_{30} \cdot \text{cong_lag}$. This component uses all 371 observations. The coefficient β_{40} directly tests the primary form of the agenda-setting hypothesis: does media coverage increase the probability that ESA listing becomes salient enough to elicit congressional activity?

Part 2 (count component) models the number of documents produced given that $cong_t > 0$, estimated from the 12 active months only:

$$\begin{aligned} cong_t \mid cong_t > 0 &\sim \text{Poisson}(\mu_i) \\ \log(\mu_i) &= \alpha_1 + \beta_{41} \cdot \text{news_lag}_i + \beta_{31} \cdot \text{cong_lag}_i \end{aligned}$$

The starting specification uses Poisson in Part 2 rather than Negative Binomial, following the recommendation of Zeileis et al. (2008). The Hurdle model already removes structural zeros through Part 1; adding NB in Part 2 would introduce a second zero-handling mechanism ($\hat{\theta}$) that risks trading off against the hurdle threshold, creating ambiguity between β_{40} and β_{41} . If residual overdispersion remains after fitting the Poisson Hurdle, the extension to NB in Part 2 can be evaluated.

The full probability mass function combines both components:

$$P(cong_t = k) = \{ 1 - P(cong_t > 0) \} k = 0 P(cong_t > 0) \cdot P_{\text{Poisson}}(k) \quad (7)$$

The two-part structure also produces two distinct coefficients for news_lag — β_{40} from Part 1 and β_{41} from Part 2 — that simultaneously answer two substantively different versions of the agenda-setting hypothesis:

- β_{40} : Does media coverage predict *whether* Congress engages with ESA listing? (the threshold question)
- β_{41} : Does media coverage predict *how much* Congress engages, given that it does? (the intensity question)

No single-component model in this progression — OLS, Poisson, Quasi-Poisson, or Negative Binomial — can answer both questions simultaneously.

The model was fitted in R (R Core Team, 2024) using the `pscl` package (Zeileis et al., 2008) with the specification:

```
hurdle(cong_t ~ news_lag + cong_lag,
       data      = df,
       dist      = "poisson",
       zero.dist = "binomial")
```

2.8 Temporal autocorrelation

Monthly observations are temporally ordered and may not be independent given only one lag of each variable. Congressional activity in month t may remain correlated with activity in month $t+2$ or beyond even after conditioning on $cong_lag$ at $t-1$, and similarly for the news series (Brandt and Williams, 2000). The current model addresses autocorrelation through the inclusion of $cong_lag$ as a covariate, following the autoregressive count model approach of Brandt and Williams (2000). A

formal test for residual autocorrelation in the Hurdle model residuals will be reported in the results. If significant residual autocorrelation is detected, extensions including a wider lag window or a correlation structure in the Part 2 component will be considered in subsequent analyses.

3 Discussion

3.1 Theoretical implications for agenda-setting

The two-part structure of the Hurdle model maps directly onto the theoretical framework of punctuated equilibrium theory (Baumgartner and Jones, 1993; Jones and Baumgartner, 2004). The logistic component in Part 1 models the punctuation — the moment at which ESA listing crosses from equilibrium (congressional silence) into active attention. The count component in Part 2 models the intensity of activity once punctuation has occurred. The media coefficient in Part 1 (β_{40}) tests whether media coverage is the mechanism that produces punctuation; the coefficient in Part 2 (β_{41}) tests whether media coverage also drives the intensity of activity after punctuation.

If both β_{40} and β_{41} are positive and significant, the finding would support a strong version of the agenda-setting hypothesis: media coverage both gets ESA listing onto the congressional agenda and drives the volume of legislative activity once it is there. If β_{40} is significant but β_{41} is not, the finding would support a threshold version: media matters for attention allocation but not for production intensity once the issue is salient. These distinctions cannot be recovered from single-component models.

This analysis also speaks to an emerging debate in political communication about whether agenda-setting operates differently for niche versus valence issues (Walgrave and Van Aelst, 2006). ESA listing is clearly a niche domain: it appears in 17.3% of news articles in our corpus but only 5.7% of congressional documents. For such issues, the threshold process modelled in Part 1 may be more theoretically important than the intensity process in Part 2, because the primary question is whether the issue makes it onto the congressional agenda at all. The Hurdle model allows both questions to be addressed without conflating them.

3.2 Methodological contributions

This paper contributes a documented model-selection protocol for rare political event count data that may be useful to researchers facing similar data structures in other policy domains. The protocol has four steps. First, characterise the outcome variable before selecting a model: document the proportion of zeros, the variance-to-mean ratio, the skewness, and the temporal clustering of non-zero observations. Second, reject OLS on structural grounds: count data with a non-negative discrete distribution cannot be appropriately modelled by a continuous normal regression. Third, evaluate the Poisson family sequentially, testing for overdispersion at each stage and documenting the specific assumption that is violated. Fourth, assess whether the zeros are structural or sampling in nature by examining whether non-zero observations cluster around identifiable events, and by comparing the observed zero frequency to the Poisson-predicted frequency at the estimated mean.

This protocol is consistent with the data exploration framework advocated by Zuur et al. (2010) and the model selection principles in Burnham and Anderson (2002), extended to the specific challenges of political count data with rare events and structural zeros.

A second methodological contribution concerns the integration of LLM-based narrative coding with statistical inference. The NPF extraction pipeline (Jones and McBeth, 2010) used in this paper represents a novel approach to operationalising theoretical constructs from political communication for large-scale quantitative analysis. The extraction of characters, plots, and morals at scale enables systematic empirical testing of narrative theory in a way that manual coding cannot support given practical resource constraints. The validation procedure described in Section 2 is an important step toward establishing the reliability of this approach, and future work should examine the conditions under which LLM-extracted narrative codes are more or less concordant with human coder judgements.

3.3 Limitations and future directions

Several limitations of the current analysis should be acknowledged. First, the congressional corpus is limited to 300 documents. While this corpus was stratified to represent the range of ESA policy activities, the monthly count series for the Species Listing category contains only 12 active months

across 30 years. This sparsity limits the precision of the count component estimates in Part 2 of the Hurdle model. Future work should extend the congressional corpus to provide a more comprehensive monthly count series.

Second, the model includes only two predictors: `news_lag` and `cong_lag`. Congressional attention to ESA listing is likely influenced by other factors that are not captured in the current specification, including election cycles, changes in committee leadership, and executive branch actions such as proposed listing decisions by the Fish and Wildlife Service. The inclusion of these covariates in both parts of the Hurdle model would strengthen causal identification and address potential confounding.

Third, the autocorrelation structure is addressed only through a single lag. As noted in Section 3.8, residual autocorrelation may remain in the Hurdle model residuals. Formal testing and, if necessary, modelling of this autocorrelation structure is an important next step.

Fourth, this paper focuses on ESA Species Listing. The other nine ESA meta-categories — including Legislative Action, Regulatory Action, Litigation, and Habitat/Recovery — each represent distinct policy processes with potentially different agenda-setting dynamics. A systematic analysis across all categories would provide a more complete picture of how media narration of ESA events relates to the congressional policy process.

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Table 1: Summary statistics for the monthly count of congressional ESA listing documents ($cong_t$), $n = 371$ months, January 1995 – December 2025.

Statistic	Value	Note
n (months)	371	January 1995 – December 2025
Months with $cong_t = 0$	359 (96.8%)	Structural zeros
Months with $cong_t > 0$	12 (3.2%)	Active months
Maximum	5	January 2025
Mean ($\bar{y}$)	0.046	Near-zero baseline
Variance ($\hat{\sigma}^2$)	0.103	Exceeds mean
Var/Mean ratio	2.25	> 1 : overdispersed
Skewness	11.27	Extreme right skew
Kurtosis	157.93	Heavy tails